



Scaling with Google Cloud Operations

This study guide covers the materials from the last course of the six-course series for the Cloud Digital Leader certification.

Certification exam focus

The Cloud Digital Leader certification exam is designed to assess knowledge of cloud literacy, common business use cases, and how cloud solutions and the capabilities of Google Cloud products support an enterprise.

Expenditures and total cost of ownership (TCO)

Moving from on-premises to the cloud shifts costs from capital expenditures (CapEx) to operational expenditures (OpEx).



CapEx: Large, upfront investments in fixed assets, which may require lead time to procure, provision, and configure.



OpEx: Pay-as-you-go expenses for services used. There are no upfront costs, but spending is variable; scaling up or down based on real-time usage.

Shifting from CapEx to an OpEx cloud model frees teams from the burden of buying and maintaining physical hardware, allowing for agility and speed.

Cloud financial governance best practices

Effective cloud financial governance requires a collaborative, cross-functional partnership between people, processes, and technology.



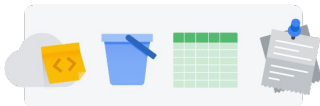
People: Identify who manages cloud costs.

Processes: Establish a routine so usage is monitored.

Technology: Use Google Cloud's tools to help gain visibility, automate cost control, and receive recommendations for optimizing resources.

Google Cloud resource hierarchy

Google Cloud's resource hierarchy organizes assets into four key levels—Resources, Projects, Folders, and the Organization node—which collectively act as a structural blueprint for your cloud footprint.



Benefits of this resource hierarchy include:

- **Granular access control:** Precise permission mapping at every level.
- **Inheritance and propagation rules:** Reduces the need to manually configure individual resources.
- **Security and compliance:** Least-privilege enforcement minimizes risk exposure.
- **Strong visibility and auditing capabilities:** Centralized tracking simplifies compliance reviews.

Controlling cloud consumption

Google Cloud provides several tools to help you control consumption:



Resource quota policies: Prevent overspending by limiting resource consumption.



Budget threshold rules: Early warning alerts for cost overruns.



Cloud Billing reports: Reactive tracking and data analysis.



Dynamic Workload Scheduler: Cost-efficient scheduling for nonurgent workloads.



Spot VMs: Deeply discounted, preemptible virtual machines.



Committed use discounts (CUDs): Cheaper rates for predictable long-term usage.





Scaling with Google Cloud Operations

Key cloud operations terms

- **Operational excellence:** The execution of a business strategy more consistently and reliably than the competition, often through automation and continuous improvement.
- **Reliability:** The probability that a system will perform its intended function without failure for a specified period of time.
- **High availability:** A system design that ensures a prearranged level of operational performance (uptime) is met during a specified period.

Modernizing with Google Cloud Observability

Cloud Monitoring



Track system performance and alert on resource health.

Cloud Logging



Centralize log data for fast incident troubleshooting.

Cloud Trace



Analyze application latency to pinpoint bottlenecks quickly.

Cloud Profiler



Optimize code execution to reduce CPU consumption.

Error Reporting



Aggregate and track application crashes for developers.

Developer operations (DevOps) and site reliability engineering (SRE)

DevOps is a collaborative approach that unites development and operations teams to accelerate and improve software delivery. SRE serves as a specific practice within this framework to guarantee the reliability and efficiency of cloud-deployed systems.

Three main concepts of SRE



Service level indicators (SLIs):

Quantitative measurements of how a service is performing. Examples include latency, error rate, and system uptime.



Service level objectives (SLOs):

The specific reliability target set for a service level indicator. It defines the goal, such as aiming for the system to be available for 99.9% of the month.



Service level agreements (SLAs):

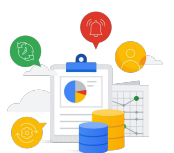
A binding contract between a service provider and a customer. It guarantees specific SLOs and outlines the penalties if the provider fails to meet them.

Measuring performance and reliability



System performance and reliability are measured using the four golden signals: Latency tracks the time it takes to service a request, while traffic measures the overall demand. Saturation monitors the utilization of resources like CPU, while errors measure the rate of failed requests, so system operates successfully.

Designing resilient infrastructure



- **Redundancy:** Eliminate single points of failure to prevent downtime.
- **Replication:** Ensure real-time data copy availability across multiple regions.
- **Scalable infrastructure:** Adapt to shifting demand by dynamically provisioning resources.
- **Backups:** Secure historical data for recovery during a disaster.

